

intermediate transformer layers together with a global [CLS] token. The DPT fusion neck reassembles and fuses these multi-scale features into a shared geometric feature map $\mathbf{G} \in \mathbb{R}^{C \times H \times W}$. We refer to the ViT backbone together with the fusion neck as the *geometric encoder*, as it produces the domain-invariant representation used by all downstream components.

From $\mathbf{G}$, two lightweight prediction heads produce complementary outputs: a depth head predicts an inverse depth map $\hat{\mathbf{d}} \in \mathbb{R}^{1 \times H \times W}$, and a normal head predicts a unit surface normal map $\hat{\mathbf{n}} \in \mathbb{R}^{3 \times H \times W}$. These heads are used only during training to provide geometric supervision; they are discarded at deployment. Predicting both modalities from a shared representation forces the encoder to learn richer geometry than either task alone would require, as depth captures the distance to surfaces while normals capture their local orientation.

For the downstream policy, the geometric encoder exports two types of features: a dense spatial feature map obtained by projecting $\mathbf{G}$ through a learned 1×1 convolution, and the [CLS] token as a global scene descriptor. During policy learning, the encoder is frozen to preserve the learned geometric inductive bias.

Model Training. The geometric encoder and prediction heads are trained jointly on synthetic RGB-depth-normal triplets rendered in Isaac Lab [21], which provides pixel-perfect ground-truth annotations. We apply color jittering and geometric perturbations with corresponding normal rotation compensation to improve robustness to visual variations. The total loss combines depth supervision, normal supervision, and a cross-task consistency term:

$$\mathcal{L}_{\text{enc}} = \underbrace{\lambda_{\text{ssi}} \mathcal{L}_{\text{ssi}} + \lambda_{\text{gm}} \mathcal{L}_{\text{gm}}}_{\text{depth}} + \underbrace{\lambda_{\text{cos}} \mathcal{L}_{\text{cos}} + \lambda_{\text{geo}} \mathcal{L}_{\text{geo}}}_{\text{normal}} + \underbrace{\lambda_{\text{dn}} \mathcal{L}_{\text{dn}}}_{\text{consistency}}. \quad (1)$$

The depth terms consist of a scale- and shift-invariant loss $\mathcal{L}_{\text{ssi}}$ [31] and a gradient matching loss $\mathcal{L}_{\text{gm}}$ that preserves depth discontinuities. The normal terms consist of a cosine loss $\mathcal{L}_{\text{cos}}$ measuring angular deviation from the ground truth and an edge-aware smoothness loss $\mathcal{L}_{\text{geo}}$ that encourages spatial coherence while respecting depth boundaries.

The central term is the depth-normal consistency loss $\mathcal{L}_{\text{dn}}$. Given a depth map and camera intrinsics $\mathbf{K}$, surface normals can be derived analytically by back-projecting pixels to 3D and computing cross products of finite differences. We exploit this geometric coupling as a self-consistency constraint:

$$\mathcal{L}_{\text{dn}} = \frac{1}{|\mathcal{M}|} \sum_{p \in \mathcal{M}} \left(1 - \hat{\mathbf{n}}_p^\top \mathbf{n}_p^{(\text{d})} \right), \quad (2)$$

where $\mathbf{n}_p^{(\text{d})}$ is the normal derived from the predicted depth at pixel p and $\mathcal{M}$ is the set of valid pixels with reliable depth. This loss requires no additional labels yet forces both heads to converge to a coherent 3D interpretation, yielding more structured and transferable features. Full details are provided in Appendix B.2.